AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 35 · STUDENT LESSON

Capstone limitations and revision

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Freeze the model. Report every case. Preserve the failure.

Evaluate a supplied candidate against its baseline and distinguish overall improvement from a structural output limit.

Keep one habit: Frozen procedure → full error ledger → output-limit graph → separate slice picture → proposed repair → executable evaluation → independent cases.

MAIN LESSON · SECTION 1

Evaluate the frozen model and keep its failures visible

> An aggregate improvement can coexist with a structural limitation.

Learning goals

I can evaluate a frozen candidate and baseline on final rows.

I can identify a distribution limitation from raw errors.

I can distinguish **revision** checks from new independent evaluation.

Retrieve capstone controls

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Capstone controlled experiments. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Documenting a network experiment. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Uncertainty and failure analysis. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Prerequisite and starting response

- After selecting rate 0.2, may you change it after viewing final scores and keep calling those scores untouched?

> **My starting explanation**

End goal

Explain capstone limitations and **revision** with a traceable toy-network case and a bounded conclusion.

▼ Earlier wording, alternatives and picture transcripts

Your learning target

Remember before reading

This week you will practise:

Evaluate a frozen candidate and baseline on final rows

Identify a distribution limitation from raw errors

Distinguish **revision** checks from new independent evaluation

Retrieve what you already know

This week's end goal

MAIN LESSON · SECTION 2

Freeze the candidate before opening the evaluation rows

> The supplied example now reveals four constructed cases, including negative inputs.

Complete frozen procedure, baseline and evaluation cases

Freeze the Week 34 candidate: max(0, a*x), initial a = 1, rate 0.2, two full-batch updates on (1,2),(2,4), no bias. The final weight is 2. The baseline is unchanged at a = 1.

Now open final rows F1=(5,10), F2=(6,12), F3=(-1,-2), F4=(-2,-4). These fictional rows deliberately include a condition absent from the positive **training** and validation **input**s.

Opening them is an evaluation event, not permission to tune on them while retaining an independent-test claim.

Raw predictions first and limited population evidence

Write raw **prediction**s for both procedures before summarising. A low overall error can hide a systematic problem in a condition. These four constructed examples do not estimate prevalence or real-world performance.

Name given and missing facts

> **List the given facts and one fact that is not supplied.**

MAIN LESSON · SECTION 3

Trace both procedures on the same four cases

> Keep signed residuals, squared errors and subset denominators explicit.

Complete raw predictions

ID · x · y · Baseline prediction · Candidate prediction

> **ID:** F1
>
> **x:** 5
>
> **y:** 10
>
> **Baseline prediction:** 5
>
> **Candidate prediction:** 10

> **ID:** F2

x: 6

y: 12

Baseline prediction: 6

Candidate prediction: 12

ID: F3

x: -1

y: -2

Baseline prediction: 0

Candidate prediction: 0

ID: F4

x: -2

y: -4

Baseline prediction: 0

Candidate prediction: 0

Every residual and squared error

Final case · Baseline residual · Baseline squared error · Candidate residual · Candidate squared error

Final case: F1

Baseline residual: -5

Baseline squared error: 25

Candidate residual: 0

Candidate squared error: 0

Final case: F2

Baseline residual: -6

Baseline squared error: 36

Candidate residual: 0

Candidate squared error: 0

Final case: F3

Baseline residual: 2

Baseline squared error: 4

Candidate residual: 2

Candidate squared error: 4

Final case: F4

Baseline residual: 4

Baseline squared error: 16

Candidate residual: 4

Candidate squared error: 16

Final case: All four rows

Baseline residual: Total squares 81

Baseline squared error: MSE 81/4 = 20.25

Candidate residual: Total squares 20

Candidate squared error: MSE 20/4 = 5

Final case: Candidate positive subset

Baseline residual: Two cases

Baseline squared error: MSE 0

Candidate residual: Candidate negative subset

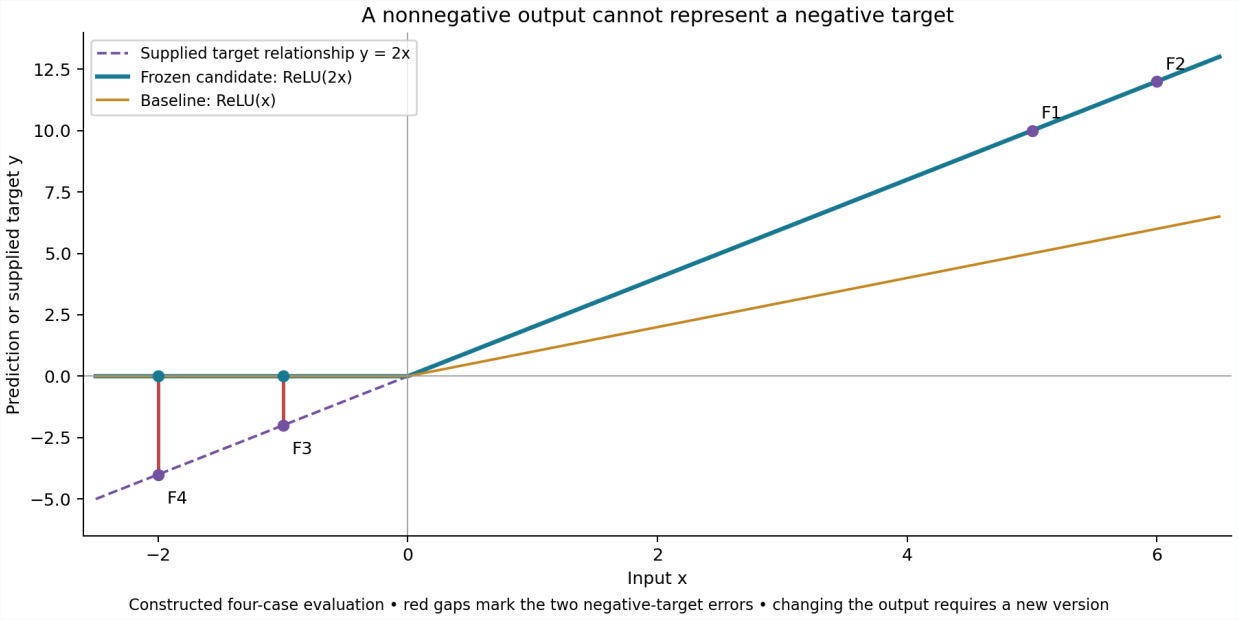
Candidate squared error: Two cases: MSE 10

Overall criterion and condition-specific scores

Baseline squared errors are 25,36,4,16, so MSE is 81/4 = 20.25. Candidate squared errors are 0,0,4,16, so MSE is 20/4 = 5. The candidate passes the strict lower-MSE final criterion on these four rows.

On the two negative-**input** rows its MSE is (4+16)/2 = 10; on the positive rows it is zero.

See the structural negative-output limit



Follow the evidence roles

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Frozen candidate a = 2	Final MSE = 5	Improves on baseline 20.25
Negative x cases	Predictions forced to zero	A structural limitation
Change output rule	Check revealed failures	New final evidence needed

A passed aggregate criterion and a known structural failure can both be true.

Explain the visual

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ **Earlier wording, alternatives and picture transcripts**

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Apply the criterion and still report the known failure

> Passing lower overall MSE does not erase negative-target errors.

Current task and response

- Compute both overall MSE values and the candidate negative-**input** MSE. Apply the strict final criterion.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

Predict the pictured exposed-case revision

- Change: Train the **revision** on all eight failed-slice evaluation patterns and rescore them.

▼ **Check after predicting**

Prediction: Treat the score as exposed-case development, not independent **evidence** of shift robustness.

Reason: The evaluation records now influenced learning and model selection.

▼ **Earlier wording, alternatives and picture transcripts**

Compare a familiar case with a changed case

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Propose a repair without overwriting the original evaluation

> Known-case checks and fresh final evidence have different roles.

Main output constraint and proposed architecture change

The ReLU **output** cannot be negative. When targets are negative, no choice of this scalar weight can produce the required negative **output**. More epochs do not remove that **output** constraint.

A proposed signed linear **output** a*x would represent negative targets, but it changes the architecture. It needs a new version, appropriate **fitting** and selection, and fresh reserved evaluation.

Preserve original results and label later checks

Running that proposal on F3 and F4 checks the revealed failures. It does not create new independent final **evidence**. Preserve the original test result and label later scores as **revision** or regression checks.

Do not replace the earlier failure report with a score from a new model on already inspected rows.

Separate picture inputs and scope

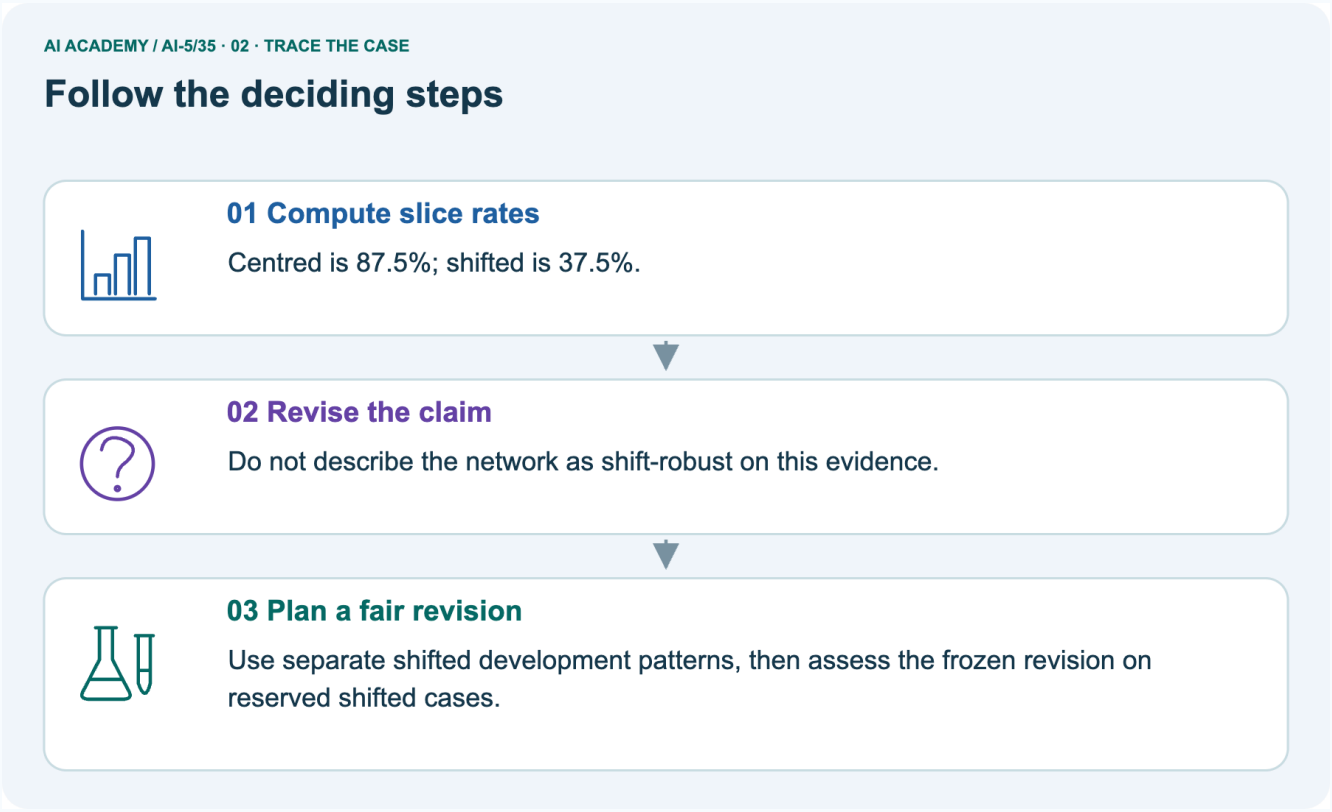
Revise a limited claim after a slice failure. A toy network works better on centred patterns than on shifted patterns, so the team narrows its initial claim.

Centred slice: 7/8 correct. Shifted slice: 3/8 correct. The network and label rubric are frozen.

The main experiment uses regression MSE; the picture is a separate supplied classification audit. Centred patterns score 7/8 = 87.5%, shifted patterns 3/8 = 37.5%. These observed counts do not prove the cause or estimate a stable future rate. Exact images and model parameters are not supplied. If all eight shifted evaluation patterns are used for a revision, rescoring them is exposed-case development evidence.

New reserved source groups are needed for an independent revised-model evaluation.

Trace observed slice rates and claim revision



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute slice rates	Centred is 87.5%; shifted is 37.5%.
2. Revise the claim	Do not describe the network as shift-robust on this evidence.
3. Plan a fair revision	Use separate shifted development patterns, then assess the frozen revision on reserved shifted cases.

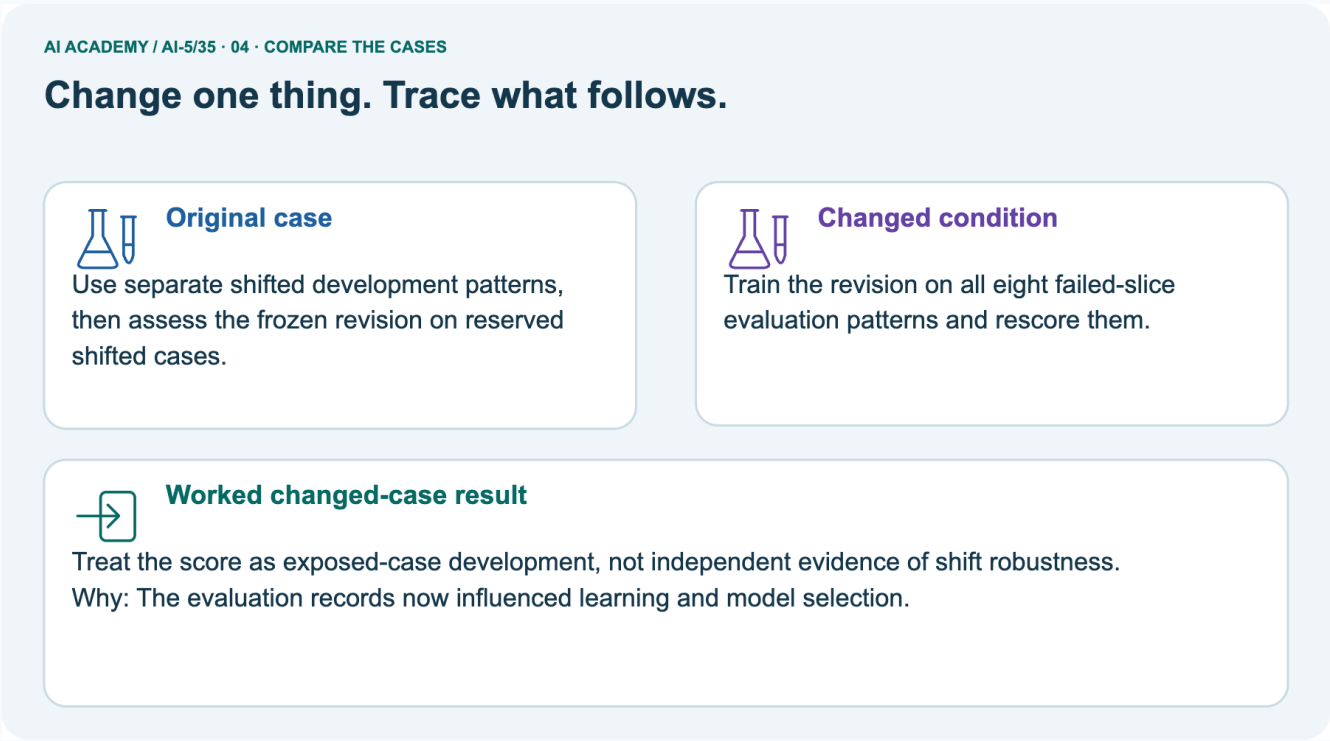
Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A limitation guides a better next experiment. **Training** on a failed evaluation example may help development, but that exposed example cannot remain the untouched **evidence** for the **revision**.

Predict training on the evaluation patterns

- Change one thing: Train the **revision** on all eight failed-slice evaluation patterns and rescore them. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain evidence role and limits

- Small slices may vary; this example diagnoses an observed weakness without inventing a population rate or cause.

- Read the labelled picture: Calculate the slice rates and explain the claim **revision**. If the team trains on the shifted evaluation patterns, what kind of **evidence** does their rerun provide?

▼ Earlier wording, alternatives and picture transcripts

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Capstone limitations and revision

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/35

Capstone limitations and revision

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute slice rates

Centred is 87.5%; shifted is 37.5%.



2. Revise the claim

Do not describe the network as shift-robust on this evidence.



3. Plan a fair revision

Use separate shifted development patterns, then assess the frozen revision on reserved shifted cases.

CHANGE ONE THING

Train the revision on all eight failed-slice evaluation patterns and rescore them.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Distinguish representational feasibility from measured generalisation

> A signed output can represent a revealed negative target without proving a broader result.

Complete supported negative-input case and guided task

- Supported example: for a = 2 on x = -3 with target -6, the ReLU **prediction** is 0, **residual** 6 and squared error 36. The signed proposal predicts -6 on this revealed case.

  This demonstrates that the alternative can represent the case; it does not establish how well a newly trained model will perform elsewhere.

Guided: calculate both predictions and errors and explain which claim the comparison supports.

Explain why extra epochs do not remove the output constraint

- Why cannot extra **training** alone make this ReLU **output** negative? Name a proposed change and the **evidence** it would require.

State one repair and the checks you would rerun.

▼ Earlier wording, alternatives and picture transcripts

Locate the earliest unsupported step

MAIN LESSON · SECTION 8

Practise residuals, subset scores and bounded claims

> Keep the model failure and the successful aggregate comparison in the same report.

Current residual task and supported case

- For F1–F4 calculate candidate **residual**s and squared errors.

Guided: calculate both predictions and errors and explain which claim the comparison supports.

Explain your prediction to a partner before comparing results.

Supported W2 record

Use workbook W2 for the connected example “Capstone limitations and **revision**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Individual criterion misconception

- Use the worked case for Capstone limitations and **revision**. Proposed explanation: “Passing a criterion means no failures remain”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Pictured revised-claim response

- Calculate the slice rates and explain the claim **revision**. If the team trains on the shifted evaluation patterns, what kind of **evidence** does their rerun provide?

> **My explanation using a pictured step and the given facts**

Reserve independent frozen a = 3

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

Practise with less help: Compare with the baseline

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Run a complete frozen-case evaluation ledger

> Preserve the original result alongside any known-case revision check.

Current overall and negative-subset task

Compute both overall MSE values and the candidate negative-**input** MSE. Apply the strict final criterion.

Complete executable frozen evaluation

```
"""Supplied frozen-case evaluation; never use this script to tune the candidate."""
final=[('F1',5,10),('F2',6,12),('F3',-1,-2),('F4',-2,-4)]
def predict(a,x):return max(0,a*x)
def evaluate(weight,rows):
    if not rows:raise ValueError('Need evaluated cases')
    ledger=[(name,x,y,predict(weight,x),(predict(weight,x)-y)**2)
            for name,x,y in rows]
    return ledger,sum(r[-1] for r in ledger)/len(ledger)
baseline,baseline_mse=evaluate(1,final)
candidate,candidate_mse=evaluate(2,final)
assert [r[-1] for r in baseline]==[25,36,4,16]
assert [r[-1] for r in candidate]==[0,0,4,16]
assert baseline_mse==20.25 and candidate_mse==5
assert candidate_mse<baseline_mse
negative=[r for r in final if r[1]<0]
positive=[r for r in final if r[1]>0]
assert evaluate(2,negative)[1]==10 and evaluate(2,positive)[1]==0
print('Frozen baseline:',baseline,'MSE:',baseline_mse)
```

```
print('Frozen candidate:',candidate,'MSE:',candidate_mse)
print('Candidate positive MSE:',evaluate(2,positive)[1], 'negative
MSE:',evaluate(2,negative)[1])
print('Strict aggregate criterion passes; negative-target representational failure
remains.')
# Proposed signed output: known-case regression check only, not fresh final evidence.
revision=[(name,2*x,(2*x-y)**2) for name,x,y in final]
assert all(error==0 for _,_,error in revision)
print('Hand-proposed signed-output check on already revealed rows:',revision)
print('Revision has not been independently fitted/selected/evaluated here.')
```

Expected outputs, evidence role and limits

1. Predict every frozen baseline and candidate output before running or tracing. The code evaluates supplied weights 1 and 2; it does not refit or retune them. Overall MSE is 20.25 versus 5, so the strict lower-MSE criterion passes on these four rows. Candidate MSE is 0 on two positive-input rows and 10 on two negative-input rows.

   The later signed-output calculation is explicitly a hand-proposed revision check on already revealed cases, not a new independent final result. If these published cases were already inspected before your own choices, record them as exposed teaching cases and reserve fresh cases for independent evaluation.

Evidence ledger and reflection

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

Predict, investigate and preserve the evidence

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: evaluate a different frozen weight and target

> Report overall improvement and the negative-input limitation together.

Complete independent four-case evaluation

A frozen ReLU candidate with a = 3 is compared with baseline a = 1 on final rows (2,6),(4,12), (-1,-3),(-2,-6). Compute both MSE values and candidate MSE on negative **input**s. Write a bounded conclusion.

First attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Feedback after trying

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Evaluate new evidence independently

Attempt this without hints

A frozen ReLU candidate with a = 3 is compared with baseline a = 1 on final rows (2,6),(4,12), (-1,-3),(-2,-6). Compute both MSE values and candidate MSE on negative **input**s. Write a bounded conclusion.

> **My independent procedure, result and limitation**
>
> _____
>
> _____
>
> _____
>
> _____

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

MAIN LESSON · SECTION 11

Preserve the failed conditions in the project record

> Keep the frozen result, revision rationale and remaining evaluation need.

Project task and artifact

For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final **evidence** pack: A frozen ReLU candidate with a = 3 is compared with baseline a = 1 on final rows (2,6),(4,12),(-1,-3),(-2,-6).

Compute both MSE values and candidate MSE on negative **input**s. Write a bounded conclusion. Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Actual run or trace and limitations

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

▼ **Earlier wording, alternatives and picture transcripts**

Evidence for your project

MAIN LESSON · SECTION 16

Finish with a narrower claim and a testable next step

> The original evaluation remains part of the record after a revision.

Retrieve and explain

- Close your notes. Explain this goal in your own words: Evaluate a frozen candidate and baseline on final rows

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Use the mathematical words precisely**

REFERENCE · SECTION 5

Reference · Use the mathematical words precisely

> Connect each term to a value or step in the current example.

Vocabulary and your explanation

Word	Meaning
Frozen procedure	A recorded procedure fixed before final evaluation
Residual	Prediction minus observed target
Revision	A documented change informed by earlier evidence
Coverage limitation	A condition or group missing or poorly represented in the evidence

Use two terms to explain a step in this week's example.

▼ Extra challenge · Explain a stronger case

REFERENCE · SECTION 10

Extra challenge · Explain a stronger case

> Choose the next challenge after the main route.

Explain,repair and extend

- For F1–F4 calculate candidate **residual**s and squared errors.
- Why cannot extra **training** alone make this ReLU **output** negative? Name a proposed change and the **evidence** it would require.
- After inspecting F1–F4, a student changes to **prediction** 2*x and reports "new final MSE 0". Explain the accurate label and the next evaluation step.

Explain → apply → challenge

> **Explain which evidence supports your conclusion and which claim remains unproved.**

▼ Reference · Check the representation and its limits

REFERENCE · SECTION 12

Reference · Check the representation and its limits

> Use after practice;reserve the independent frozen a = 3 evaluation for the first individual attempt.

Current explanation and response

- A **revision** record names the observed failure, proposed **mechanism**, changed rule, preserved earlier results and new evaluation plan. Reserve fresh cases spanning the relevant **input** conditions.

  Include regression checks for earlier successes, and keep independent evaluation separate. Record any cases still missing from coverage.

- After inspecting F1–F4, a student changes to **prediction** 2*x and reports “new final MSE 0”. Explain the accurate label and the next evaluation step.

Exit: write one sentence reporting the original final result and one sentence describing what remains unverified after the proposed architecture change.

My limited claim and a useful next check

REFERENCE · SECTION 13

Extra practice · Revisit the complete trace

> Predict before reading the supplied steps.

Complete trace and evidence

Complete evidence record

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Frozen candidate a = 2	Final MSE = 5	Improves on baseline 20.25
Negative x cases	Predictions forced to zero	A structural limitation
Change output rule	Check revealed failures	New final evidence needed

A passed aggregate criterion and a known structural failure can both be true.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

▼ **Reference · The earlier route checklist**

REFERENCE · SECTION 15

Reference · The earlier route checklist

> Use one main route rather than repeating the lesson.

Targets,route and first prediction

Start with this question: Revise a limited claim after a slice failure. A toy network works better on centred patterns than on shifted patterns, so the team narrows its initial claim.

By the end, I can	How I will show it
Evaluate a frozen candidate and baseline on final rows	Explain the deciding step, show a trace or test, and state one limit.
Identify a distribution limitation from raw errors	Explain the deciding step, show a trace or test, and state one limit.
Distinguish revision checks from new independent evaluation	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 17

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Revise a limited claim after a slice failure. A toy network works better on centred patterns than on shifted patterns, so the team narrows its initial claim.
- Which labelled step matches this detail? Use separate shifted development patterns, then assess the frozen **revision** on reserved shifted cases.
- 1. Compute slice rates
- 2. Revise the claim
- 3. Plan a fair **revision**
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Plan a fair **revision** Use separate shifted development patterns, then assess the frozen **revision** on reserved shifted cases.

A limitation guides a better next experiment. **Training** on a failed evaluation example may help development, but that exposed example cannot remain the untouched **evidence** for the **revision**.

Changed-case question and optional hint

- Changed-case practice: Train the **revision** on all eight failed-slice evaluation patterns and rescore them. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Treat the score as exposed-case development, not independent **evidence** of shift robustness.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Treat the score as exposed-case development, not independent **evidence** of shift robustness. The evaluation records now influenced learning and model selection.

Repair and retain the record

Review Capstone controlled experiments (ai-5/34). Goal: Trace matched two-update learning-rate experiments. Short practice: Compute the two correct counts and identify the changed case.

Why would changing B's **training** data weaken the conclusion about its added connection?. Return to this week and explain what you changed.

Review Documenting a network experiment (ai-5/32). Goal: Record a network experiment with enough detail to replay its calculations. Short practice: What additional **evidence** is needed for a final performance claim?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Uncertainty and failure analysis (ai-5/29). Goal: Distinguish a predicted class, confidence score and observed correctness. Short practice: What is C's positive-class probability and chosen-label confidence?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

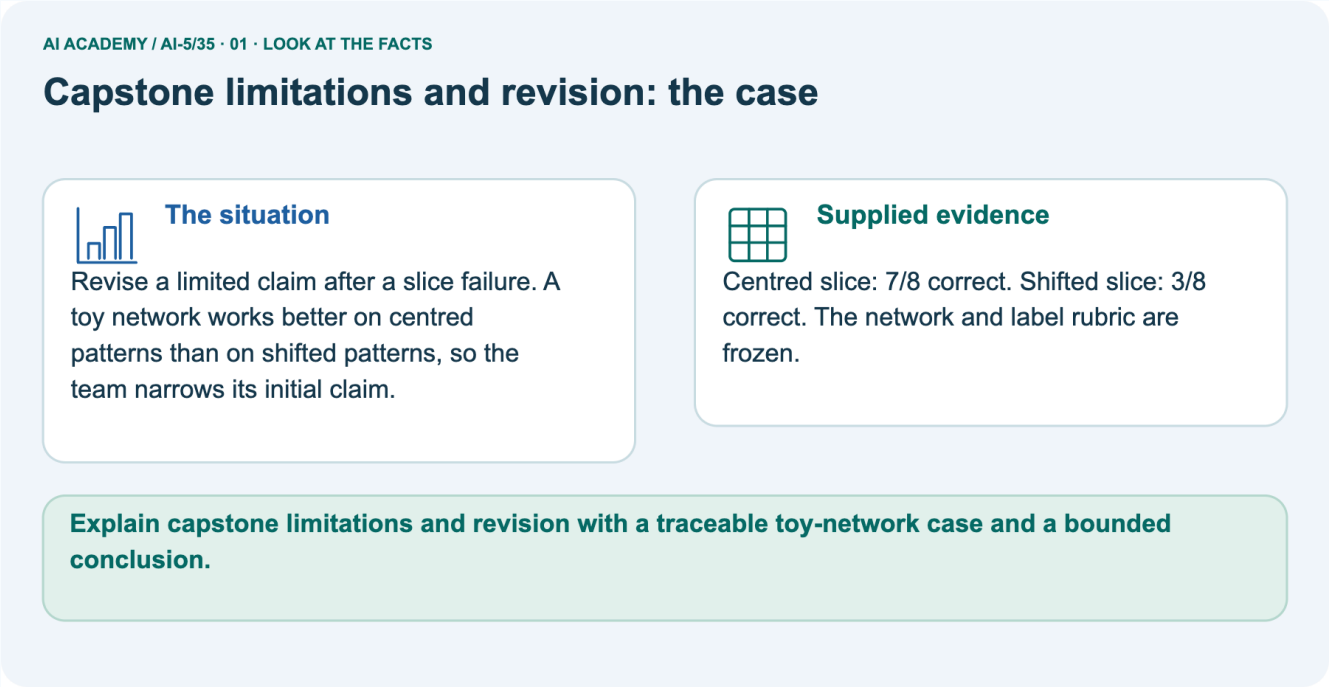
▼ Reference · Six-picture concept journey

REFERENCE · SECTION 18

Reference · Six-picture concept journey

> Choose the picture that supports the current explanation.

01. Capstone limitations and revision: the case



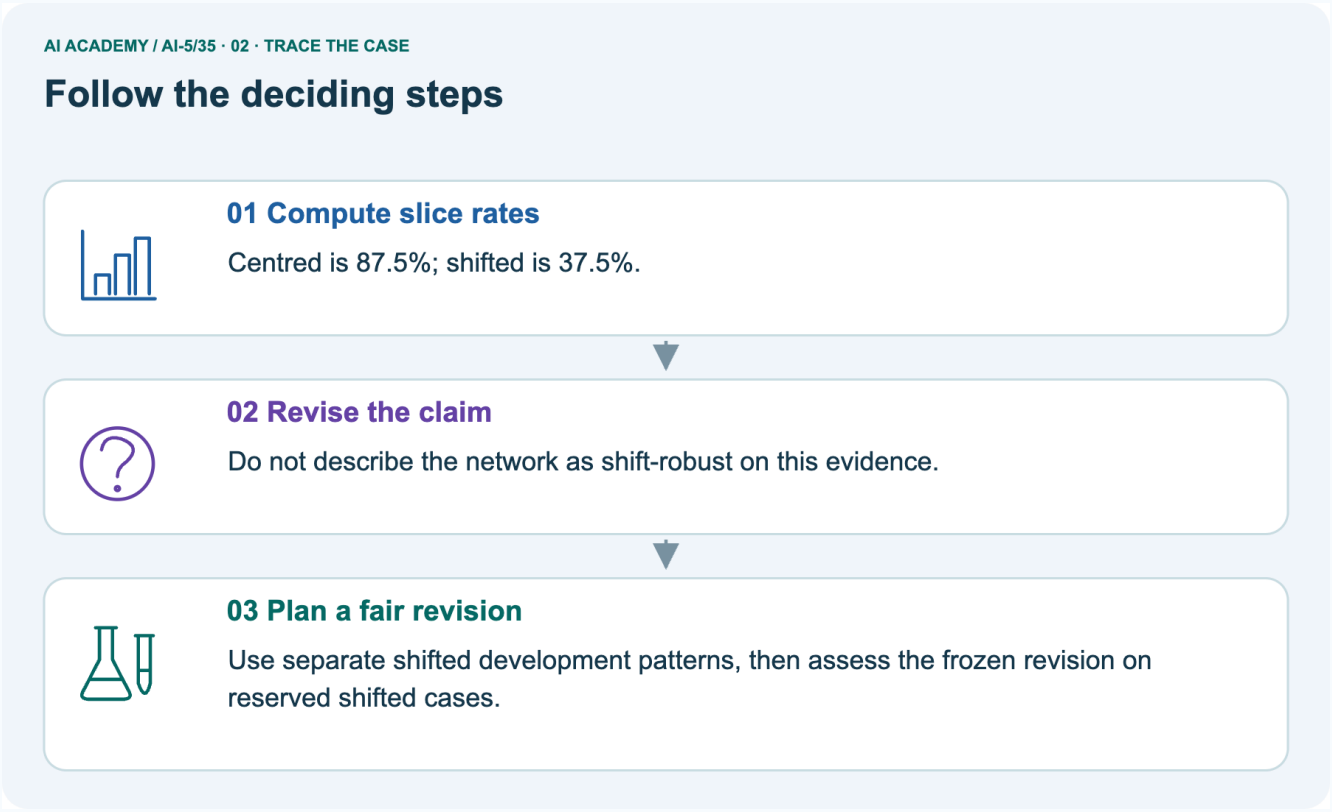
Follow the labelled picture with the matching explanation. Open the vector diagram

Revise a limited claim after a slice failure. A toy network works better on centred patterns than on shifted patterns, so the team narrows its initial claim. Supplied facts: Centred slice: 7/8 correct.

Shifted slice: 3/8 correct. The network and label rubric are frozen.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

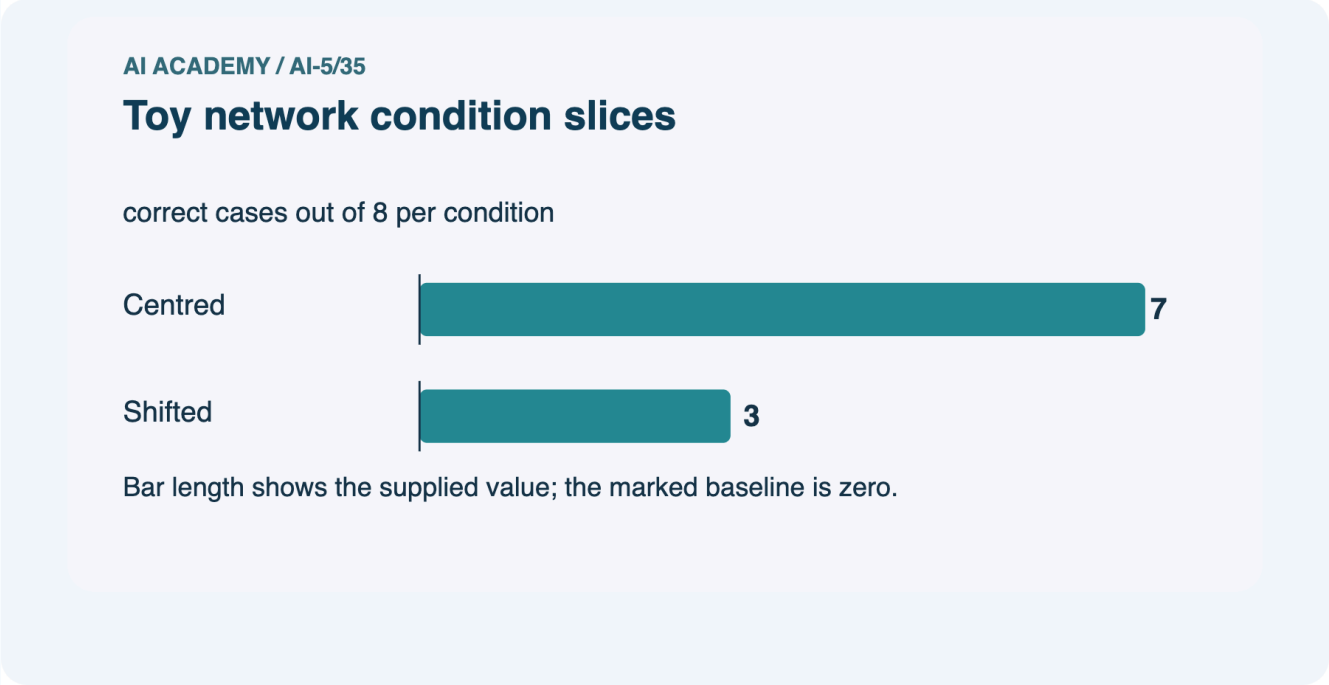


Follow the labelled picture with the matching explanation. Open the vector diagram

Compute slice rates: Centred is 87.5%; shifted is 37.5%. → Revise the claim: Do not describe the network as shift-robust on this **evidence**. → Plan a fair **revision**: Use separate shifted development patterns, then assess the frozen **revision** on reserved shifted cases.

Point to each numbered stage and explain which recorded **input** it uses.

03. Toy network condition slices

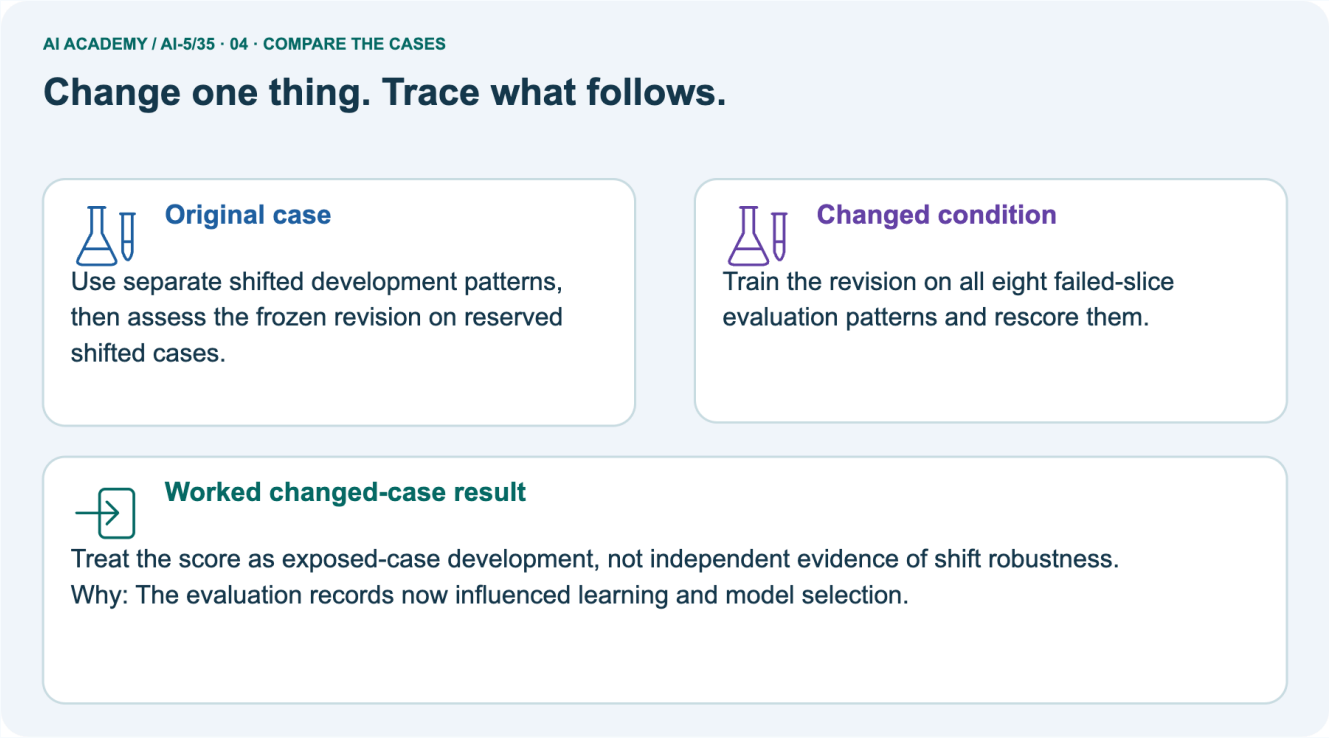


Follow the labelled picture with the matching explanation. Open the vector diagram

Toy network condition slices. Units: correct cases out of 8 per condition. Centred: 7. Shifted: 3.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



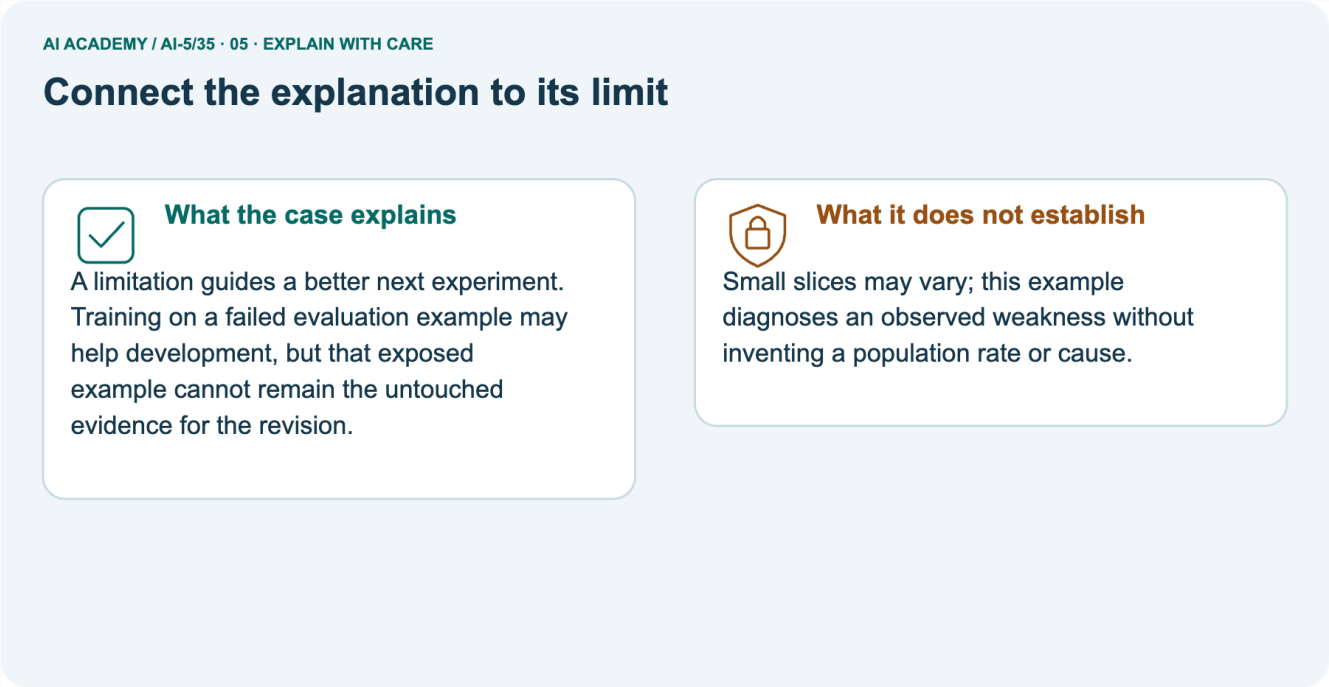
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Use separate shifted development patterns, then assess the frozen **revision** on reserved shifted cases. Changed condition: Train the **revision** on all eight failed-slice evaluation patterns and rescore them.

Expected result: Treat the score as exposed-case development, not independent **evidence** of shift robustness. Reason: The evaluation records now influenced learning and model selection.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



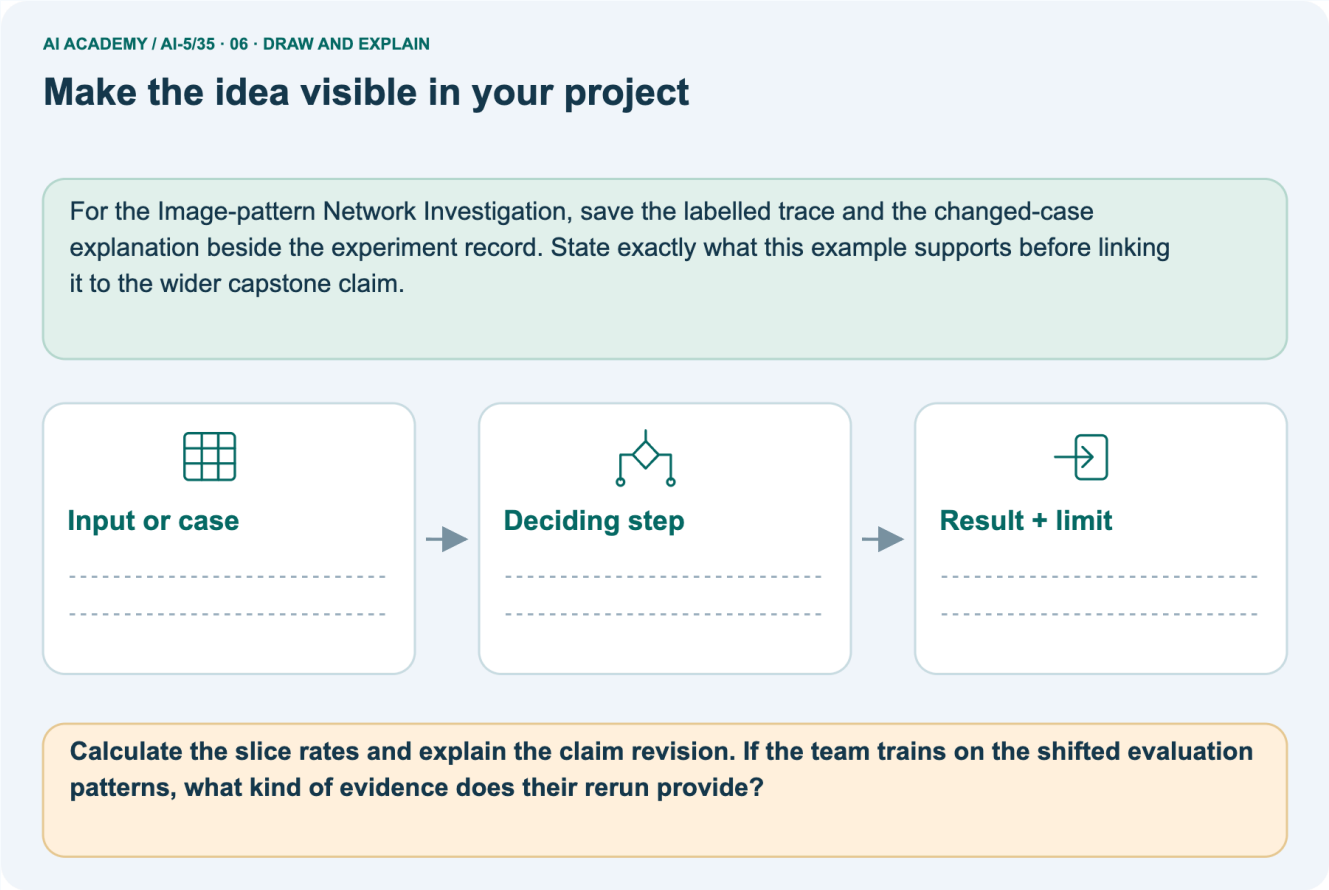
Follow the labelled picture with the matching explanation. Open the vector diagram

A limitation guides a better next experiment. **Training** on a failed evaluation example may help development, but that exposed example cannot remain the untouched **evidence** for the **revision**.

Limit: Small slices may vary; this example diagnoses an observed weakness without inventing a population rate or cause.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the slice rates and explain the claim **revision**.

If the team trains on the shifted evaluation patterns, what kind of **evidence** does their rerun provide?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Capstone limitations and revision: the case

02. Follow the deciding steps

03. Toy network condition slices

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.